

AirSense

Campus Pilot Initiative

Rooftop Air Quality Monitoring, Intelligent Forecasting and Campus Decision Support

A Proposal for Budget Approval, Hardware Deployment and Pilot Validation

SUBMITTED TO

Ms. Saba Ahson

Head of Institute | Beaconhouse International College, Islamabad

Ms. Sahifa Alam

Head of CSSE / AI Department | Beaconhouse International College,
Islamabad

SUBMITTED BY

Munim Qureshi

AirSense Project Team | Beaconhouse International College, Islamabad

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PKR 20,000

Budget Requested

3 Months

Pilot Duration

2 Sites

Islamabad + Karachi

4 Horizons

1h / 3h / 6h / 24h Forecasts

1. Executive Summary

AirSense is a student-developed air quality monitoring and intelligent forecasting platform built by a team at Beaconhouse International College (BIC), Islamabad. The system was conceived and refined during participation in a competitive international academic innovation programme and has since evolved into a full-stack, AI-powered environmental decision support platform.

This proposal requests the approval of a budget of PKR 20,000 to fund the Phase 1 campus pilot at BIC Islamabad. The pilot involves the installation of a low-cost rooftop sensor unit on the BIC campus, the collection of three months of continuous air quality and meteorological data, and the training and validation of machine learning forecasting models against that real-world dataset. A parallel pilot deployment is being coordinated at a second BIC Campus in Karachi, establishing a two-city data network from the outset.

The PKR 20,000 budget covers all hardware procurement, rooftop installation, weatherproofing, and a contingency reserve. No cloud, subscription, or recurring costs are required during the pilot period. The institute's total exposure is a single, one-time capital expenditure.

At the close of the three-month pilot, BIC will be in possession of a validated local air quality dataset, trained ML forecasting models calibrated to Islamabad conditions, a tested operational decision framework, and a comprehensive pilot completion report. These are tangible, lasting academic and institutional assets produced from a modest initial investment.

2. Project Background and Origin

2.1 How AirSense Began

AirSense was developed by the BIC student team during an international academic innovation and business challenge programme that brought together students from multiple institutions and countries. The project was selected for its novel approach to environmental data: rather than building another monitoring dashboard, the team designed a system that operates at the decision layer, translating air quality forecasts into plain-language operational recommendations that institutions and businesses can act upon.

The initial prototype demonstrated three core capabilities: real-time PM2.5 monitoring, multi-horizon AI-powered forecasting (1-hour through 24-hour ahead), and an operational decision framework that maps forecast outputs to specific actions. The system was trained on publicly available datasets for early validation, but the team identified from the outset that real-world deployment and locally collected data were essential for the system to achieve meaningful predictive accuracy.

2.2 Why the Campus Pilot Is the Right Next Step

Every scalable system requires a controlled first environment in which to identify errors, validate assumptions, and stress-test components before broader deployment. The BIC campus provides exactly this: a defined geographic environment, available infrastructure, a community of stakeholders who are direct beneficiaries of the data, and institutional accountability that ensures rigour throughout the process.

The campus pilot converts AirSense from a validated concept into a functioning, data-producing system. The dataset collected at BIC Islamabad will be the first genuinely local, continuously collected PM2.5 time series from this institution, and the models trained on it will reflect the actual meteorological and human-activity patterns of the campus environment. This is a meaningful scientific contribution in its own right.

2.3 The Problem This Project Addresses

Air quality degradation is among the most consequential environmental challenges in Pakistan's urban centres. Fine particulate matter, classified as PM2.5, comprises particles smaller than 2.5 micrometres in diameter. At sustained concentrations, it is associated with respiratory illness, cardiovascular disease, and reduced cognitive function. The Islamabad-Rawalpindi region experiences significant seasonal air quality deterioration, yet localised, continuously updated data remains scarce.

Institutions currently make decisions about outdoor activities, ventilation management, and campus operations without access to real-time, locally relevant environmental data. AirSense is designed to fill this gap, first for the BIC campus itself, and subsequently for a broader range of institutional and commercial users.

3. The AirSense System

3.1 System Architecture

AirSense operates across four functional layers, each building on the last to convert raw sensor readings into actionable intelligence.

Layer	Component	What It Does
Sensing	PMS7003 + BME280 + Rain Sensor	Measures PM1, PM2.5, and PM10 particulate concentrations alongside temperature, humidity, barometric pressure, and rainfall events at the rooftop level.
Control and Logging	ESP32 Microcontroller + MicroSD	Processes sensor data, applies timestamps, logs continuously to an onboard SD card, and transmits aggregated hourly records to a campus workstation via Wi-Fi.
ML Forecasting Engine	Random Forest / XGBoost / LightGBM	Trains on the collected dataset to predict PM2.5 concentrations at 1-hour, 3-hour, 6-hour, and 24-hour horizons. Walk-forward validation ensures forecasts reflect real operational performance.

Decision Layer	Operational Playbook + AI Advisor	Maps forecast outputs to plain-language operational recommendations. Tested in pilot mode against observed conditions to assess accuracy and practical utility.
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3.2 What Makes AirSense Different

Most air quality systems deliver data. AirSense delivers decisions. The distinction is intentional and architectural. The system does not ask institutions to interpret a PM2.5 reading and determine an appropriate response. Instead, it processes that reading alongside forecast data and says: adjust ventilation settings, move outdoor activity indoors, or reduce physical exertion for the next three hours. This decision-layer positioning is the core innovation of the platform.

A second distinguishing feature is the system's portability. Once the modelling pipeline is trained and validated at a given location, deploying to a new city requires retraining on local data rather than rebuilding the architecture. The BIC Islamabad pilot directly produces a validated blueprint for replication.

3.3 Forecasting Feature Set

The ML models consume the following inputs to generate PM2.5 forecasts. Features are grouped below by source and role.

Group	Feature	Source	Role in Forecasting
Target	PM2.5 Concentration	PMS7003	Primary variable to forecast across all four horizons.
Lag Features	PM2.5, PM10, PM1 (t-1 to t-24)	PMS7003	Captures persistence and autocorrelation. Most predictive feature group at short horizons.
Meteorology	Temperature, Humidity, Pressure	BME280	Atmospheric mixing and hygroscopic particle growth drivers.
Meteorology	Rainfall Event Flag	Rain Sensor	Wet deposition events sharply reduce PM2.5. Flagged as binary feature.
Meteorology	Wind Speed and Direction	Public Weather API	Primary dispersion driver. Sourced from open meteorological feed.
Temporal	Hour, Day of Week, Rolling Means	Software-derived	Diurnal patterns, weekday/weekend activity, and smoothed trend signals.
Context	Academic Activity Flag	Manual log	Distinguishes term-time patterns from break periods for model calibration.

4. Campus Pilot Plan

4.1 Pilot Objectives

1. Deploy a weatherproofed PM2.5 sensor assembly on the BIC Islamabad rooftop and verify stable, continuous live operation.
2. Collect a minimum three-month hourly dataset comprising PM1, PM2.5, PM10, temperature, humidity, barometric pressure, and rain-event indicators.

3. Apply feature engineering to prepare the dataset for machine learning: lag variables, rolling averages, time-of-day flags, academic calendar flags, and wind-direction bins.
4. Train and evaluate tree-based forecasting models across four prediction horizons using walk-forward time-series validation.
5. Test the operational decision-making layer in pilot mode, assessing the accuracy and utility of triggered recommendations against observed conditions.
6. Deliver a complete pilot report to BIC management covering model performance metrics, data quality findings, infrastructure lessons, and a Phase 2 roadmap.

4.2 Setup and Deployment Steps

The deployment follows a phased approach from approval to live operation. Each phase has clear entry conditions and completion criteria.

Phase	Timeframe	Activities
Phase 0	Days 1-7	Hardware Procurement - Place orders with verified vendors: Embeded Studio, Digilog.pk, Electrobes, A.E Solution, Daraz.pk, and local suppliers. - Confirm stock, delivery estimates, and model specifications before payment. Capture order confirmations. - Obtain local radiation shield quotation and procure weatherproof extension cable from Islamabad suppliers.
Phase 1	Days 6-9	Bench Assembly and Testing - Assemble all components on the workbench: wire PMS7003 and BME280 to ESP32 via I2C/UART; connect SD module via SPI; wire power supply. - Flash ESP32 firmware. Verify sensor readings, SD card write operations, and Wi-Fi transmission to the campus workstation. - Confirm all components are operational before any outdoor installation. Photograph assembly and record test readings.
Phase 2	Day 9-10	Rooftop Installation and Commissioning - Mount the weatherproofed enclosure at the agreed rooftop location using stainless hose clamps. Route and seal the power cable. - Position campus Wi-Fi extender for rooftop coverage. Verify signal strength and confirm live data transmission to workstation. - Conduct end-to-end data flow check: sensor readings appearing in SD log and on workstation. System is live. Pilot clock starts.
Phase 3	Month 1-3	Continuous Data Collection (Autonomous) - Sensor collects readings every 5-10 minutes. Data logged locally to SD card and transmitted via Wi-Fi. No daily intervention required. - System operates autonomously through academic breaks. Summer recess data is scientifically valuable as a low-activity baseline. - Weekly data quality checks. Public weather API feed (wind speed/direction) synchronised to hourly dataset.
Phase 4	Month 2-3	Feature Engineering and Model Training - Clean and quality-control the dataset. Remove outliers, flag gaps, verify SD and Wi-Fi logs for consistency. - Engineer lag features, rolling means, temporal flags, and event indicators. Merge with public wind data feed.

		Train Random Forest, XGBoost, and LightGBM models per forecast horizon. Evaluate using walk-forward validation. Apply SHAP for interpretability.
Phase 5	End Month 3	Reporting and Closeout - Test decision layer in pilot mode. Log triggered recommendations and compare against observed conditions. - Compile pilot completion report: dataset summary, model RMSE/MAE/R2 per horizon, decision framework accuracy, infrastructure notes. - Submit report to BIC management. Archive dataset and trained models. Plan Phase 2 based on findings.

The pilot is designed for autonomous, low-maintenance operation. Once installed, the sensor unit requires no daily human intervention. Data collection continues uninterrupted through holidays and academic breaks, accumulating a dataset that grows more statistically robust with every week of operation.

5. Karachi Campus Collaboration

5.1 Two-City Pilot Network

In parallel with the Islamabad deployment, a second pilot is being coordinated at a Beaconhouse campus in Karachi. A dedicated team member has joined the AirSense project specifically to lead the Karachi deployment, using the same sensor hardware specification, firmware, and data pipeline established at BIC Islamabad.

This two-site structure is not accidental. Karachi and Islamabad present substantially different air quality profiles driven by distinct topographies, traffic patterns, industrial presence, and sea-proximity effects. Collecting simultaneous data from both cities during the same three-month window produces a multi-city training dataset that accelerates model generalisation and validates the portability of the AirSense architecture.

BIC Islamabad	BIC Karachi
- Primary pilot site. Full team present for assembly, commissioning, and ongoing monitoring. - Dataset anchors ML model training. Results used as performance benchmark. - Inland topography, cooler winters, distinct traffic and dust patterns.	- Parallel deployment led by a dedicated Karachi-based team member joining specifically for this expansion. - Identical hardware stack and data pipeline. Enables direct inter-city comparison. - Coastal climate, different seasonal pollution drivers, industrial proximity data.

5.2 Why This Matters for the Pilot

A two-city simultaneous dataset significantly increases the scientific value of the pilot. Models trained on data from a single location can overfit to its specific microclimate. By incorporating Karachi data, the team can test whether the same model architecture generalises across Pakistani cities or

whether city-specific tuning is required. This finding directly informs the expansion strategy for Phase 3 and beyond.

From BIC Islamabad's perspective, the Karachi collaboration also validates the scalability of the initiative. The fact that a second institution is adopting the same system demonstrates that the design is reproducible and the methodology is sound.

6. Benefits to Beaconhouse International College

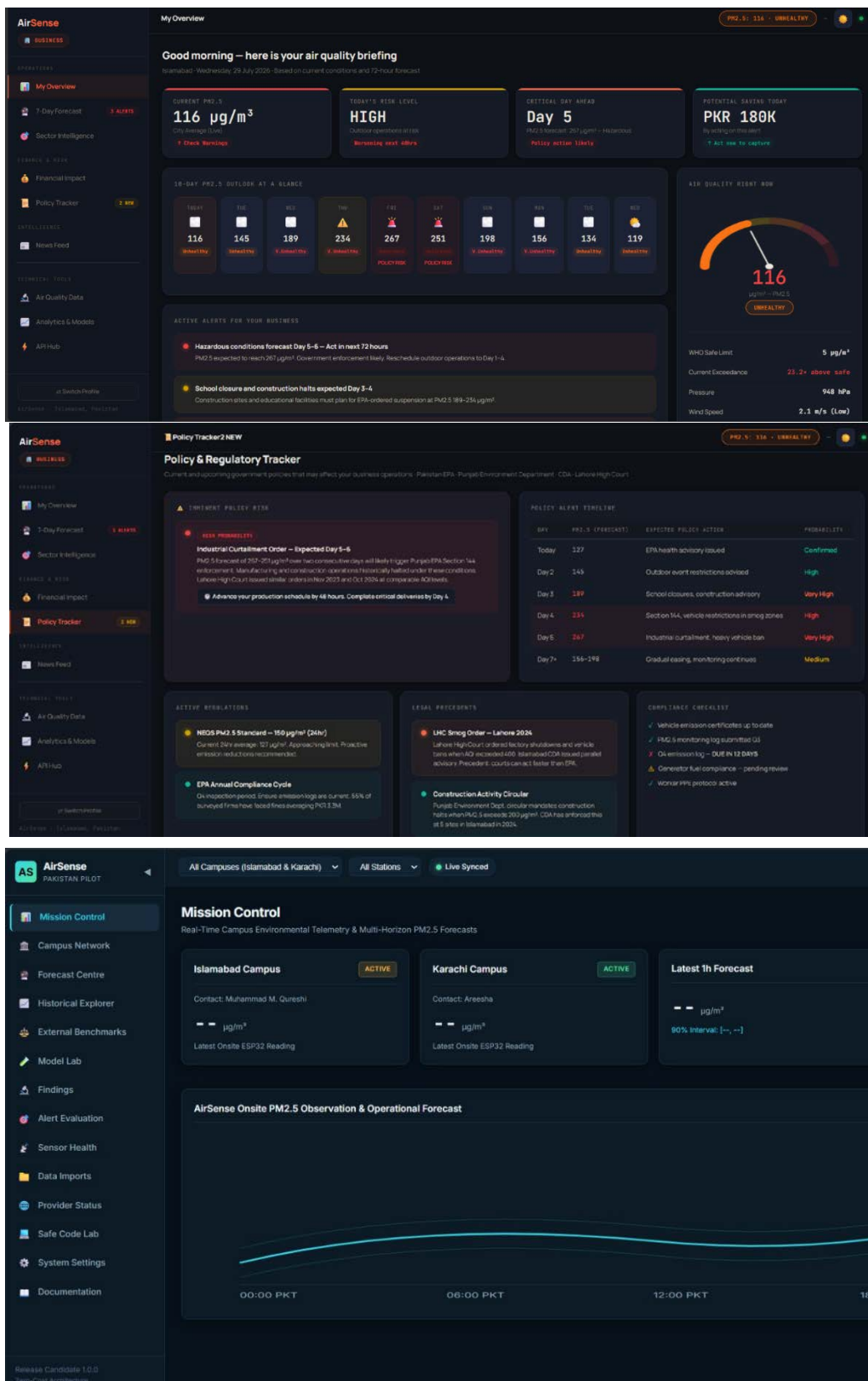
Benefit Area	What BIC Gains
Campus Health Awareness	Real-time, locally measured PM2.5 data specific to the BIC rooftop environment. Provides a factual basis for decisions about outdoor activities, ventilation, and physical education scheduling.
Academic Research Asset	Three months of continuously collected, quality-controlled environmental data from a campus deployment. This dataset has independent academic value and can support student research projects across the AI, Computer Science, and Business departments.
Student Innovation Showcase	A live, functional AI and IoT based project deployed on campus demonstrates the applied capability of BIC students. The initiative reflects directly on the quality of the AI and CSSE programmes and provides a concrete showcase for prospective students and institutional partners.
Institutional Positioning	BIC Islamabad becomes an early adopter of campus-level AI-driven environmental monitoring in Pakistan. The institute gains recognition as an institution that supports applied student innovation with tangible, deployed outcomes.
AI Department Credibility	A functioning real-world ML deployment pipeline, trained on campus-generated data, is a strong academic credential for the AI and CSSE department. It demonstrates that the department produces graduates capable of building and deploying systems, not only studying them.
Pilot Report Ownership	All data, trained models, code, and documentation produced during the pilot are submitted to BIC. The institute retains full ownership of these assets and may use them for future research, academic reporting, or further institutional deployment.

7. Dashboard Overview

The AirSense dashboard is the primary interface through which forecast data and operational recommendations are presented to users. During the campus pilot, a lightweight version of the dashboard is operated on a campus workstation to display live sensor data, forecast outputs, and system status. The full production dashboard has been developed and tested by the team as part of the broader platform build.

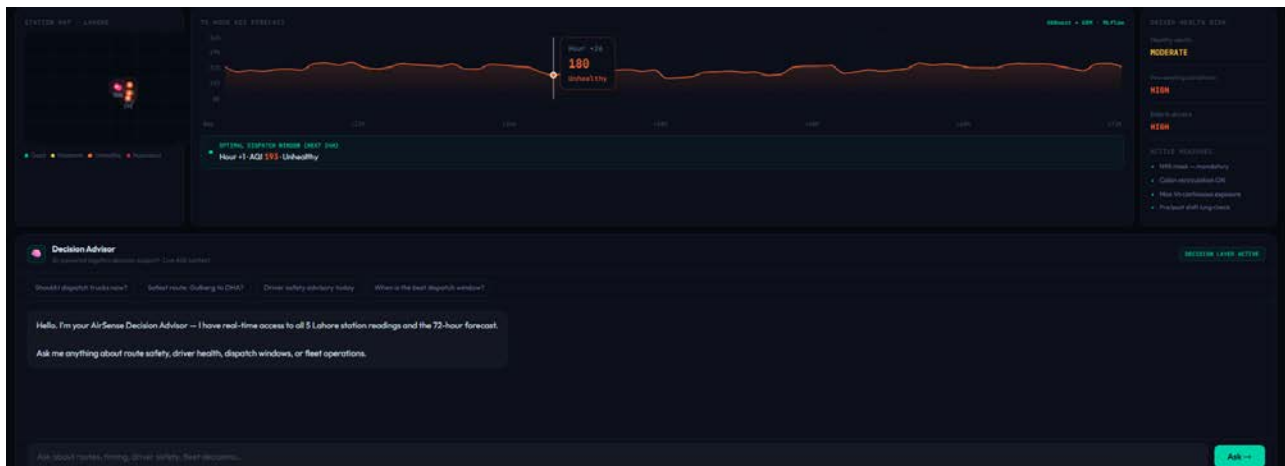
7.1 Main Dashboard View

The main dashboard presents the current AQI reading, real-time PM2.5 concentration, a colour-coded air quality status indicator, and the station status panel. Navigation tabs at the top allow the user to switch between the live view, forecast panel, health risk panel, and AI advisor.



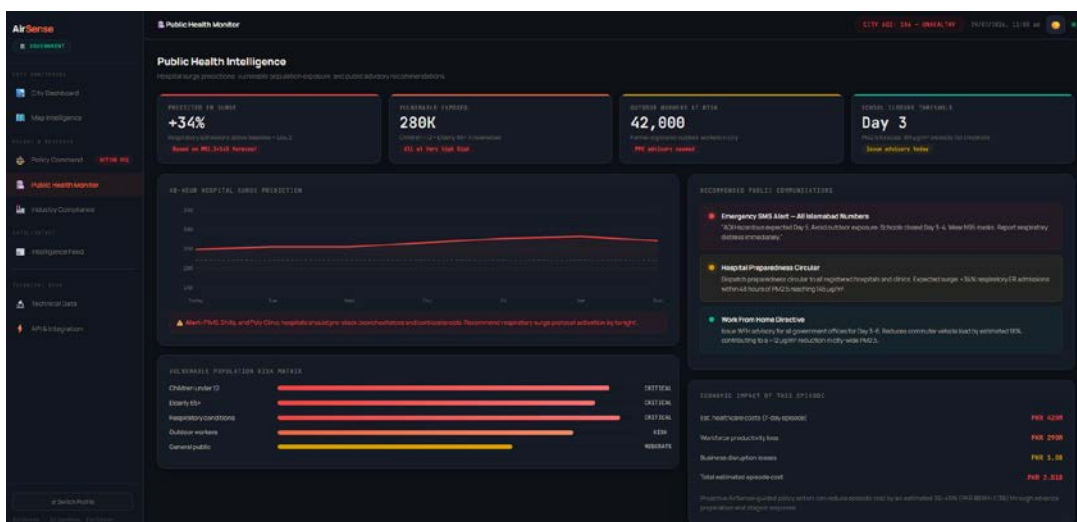
7.2 AI Decision Advisor

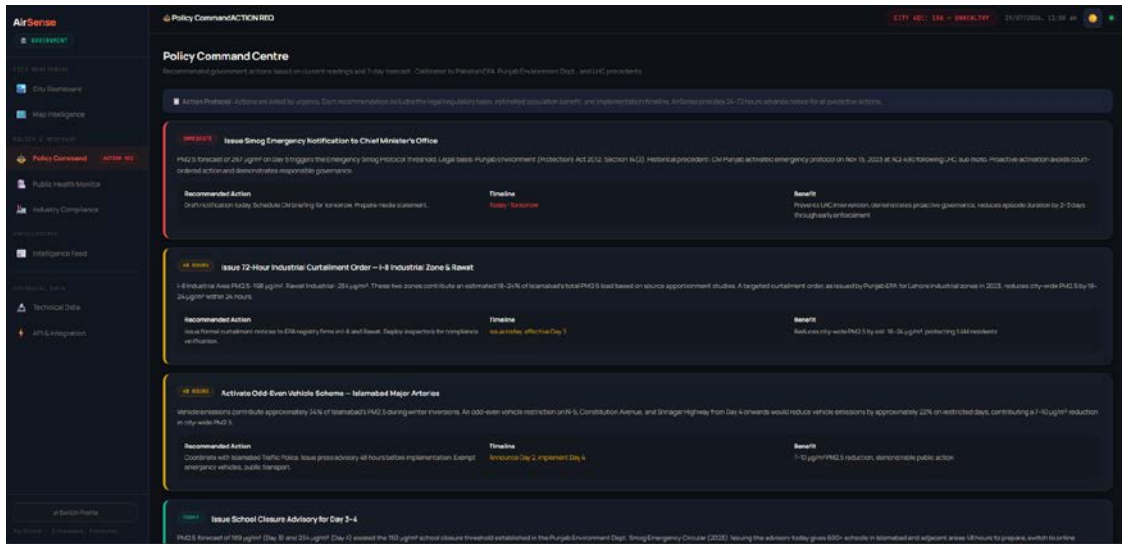
The AI Decision Advisor accepts natural-language queries from the user and responds with forecast-informed operational guidance. It ingests live sensor data and forecast outputs automatically, enabling queries such as: Is it safe to hold outdoor activities tomorrow morning? or What ventilation settings are appropriate for the next six hours? The advisor is powered by a large language model integrated directly into the dashboard pipeline.



7.3 Health Risk Panel and Dispatch Window Calculator

The health risk panel classifies current and forecast conditions into risk tiers for different user groups, including staff, students, and outdoor workers. The dispatch window calculator identifies optimal low-pollution periods for time-sensitive operational decisions.





8. Future Vision and Expansion Path

8.1 From Campus Pilot to Commercial Platform

The campus pilot is Phase 1 of a multi-stage roadmap. Each phase is designed to generate the assets, validated models, and institutional credibility needed to support the next. The table below sets out the progression from the BIC pilot to a city-level and commercially deployed system.

Phase	Stage	Description	Key Output
1	Campus Pilot	Deploy rooftop sensor at BIC Islamabad and Karachi campus. Collect three months of data. Train and validate ML forecasting models.	Validated local dataset, trained models, tested decision framework, pilot completion report.
2	Model Refinement	Retrain models on validated Islamabad and Karachi data. Improve feature engineering and decision framework based on pilot findings. Expand to additional BIC campus locations.	Higher-accuracy models. Multi-campus dataset. Decision framework version 2.
3	Commercial SaaS Launch	Deploy the platform as a subscription service targeting logistics operators, construction firms, large campuses, and retail chains in Lahore, Karachi, and Islamabad. Live cloud data pipeline. API integrations with eight data sources.	Revenue-generating SaaS product. Full cloud infrastructure. Operational decision support for paying business clients.

8.2 Operational Decision Making for Businesses

The commercial application of AirSense is positioned at the intersection of environmental data and operational management. The platform does not sell air quality readings; it sells decisions. The following table illustrates how forecast outputs translate to specific operational actions across key target sectors.

Sector	Trigger Condition	Operational Action Recommended
Logistics and Delivery	High PM2.5 forecast for morning peak	Shift loading, unloading, and route dispatch to lower-exposure windows. Alert drivers. Reschedule high-exertion tasks.
Campuses and Institutions	PM2.5 rising in next 3 to 6 hours	Increase indoor filtration. Reschedule outdoor activities. Reduce fresh-air intake through ventilation systems.
Construction and Site Work	High PM2.5 and low wind dispersion	Reduce dust-generating activity. Increase wet suppression. Monitor worker exposure. Pause outdoor operations if thresholds exceeded.
Retail and Hospitality	Poor air quality episode forecast	Adjust HVAC settings. Display indoor air quality messaging to customers. Protect outdoor staff with appropriate rest schedules.
ESG and HSE Reporting	Repeated exceedance of health thresholds	Generate automated alert logs for incident records, mitigation evidence, and business continuity planning documentation.

8.3 The BIC Pilot as the Foundation

Every element of the commercial and regional expansion path described above depends on the existence of a validated, locally trained model. The BIC campus pilot produces exactly this. The dataset collected during the three-month pilot is the training foundation for the Islamabad-calibrated version of AirSense. Without real local data, the system operates on assumptions. With it, the system operates on evidence.

Approving this pilot does not merely fund a student project. It funds the data collection and model validation that makes the entire subsequent roadmap achievable. The PKR 20,000 investment at this stage is the enabling condition for everything that follows.

9. Budget Request

9.1 What PKR 20,000 Covers

The requested budget of PKR 20,000 is a one-time capital expenditure. It covers all hardware required to deploy a complete, weatherproofed, continuously operating rooftop sensor unit. No cloud infrastructure, no subscriptions, and no recurring costs are required during the pilot period. The breakdown below reflects updated vendor pricing as of July 2026.

#	Component	Vendor	Qty	Cost (PKR)
1	PMS7003 PM1/PM2.5/PM10 Laser Sensor	Embedded Studio	1	4,200
2	ESP32 WROOM-32D Development Board	Digilog.pk	1	1,160
3	BME280 Temperature / Humidity / Pressure Sensor	Electrobes	1	850

4	Arduino MicroSD Card Reader Module	Electrobes	1	150
5	Samsung EVO Plus 16 GB MicroSD Card	Daraz.pk	1	1,450
6	Waterproof IP65-rated Electrical Enclosure	A.E Solution	1	750
7	Jumper Wire Kit, 120 pcs	Daraz.pk	1	440
8	5V 2A AC/DC Power Adapter	Electronics Hub	1	170
9	Cable Ties, 100-pack	Daraz.pk	1	120
10	2-inch Stainless-Steel Hose Clamps, 2 pcs	DreamsMart.pk	1	60
11	Raindrop Detection Sensor Module A	Digilog.pk	1	160
12	Rain Drop Moisture Detection Sensor Module B	Electrobes	1	150
13	UNI-T UT363 Digital Anemometer (wind validation)	Electrobes	1	4,250
14	Clopal 10 m Heavy-Duty Extension Lead	Clopal Online	1	3,295
15	GMSA RTV Silicone Weatherproof Sealant	Expert Tools World	1	390
16	Mounting Nuts and Bolts M3-M6 (8 pcs)	Multan Electronics	8	192
17	Radiation Shield (local fabrication / sourcing)	Local Islamabad supplier	1	TBD
Hardware Subtotal (Items 1-16, excluding radiation shield)				PKR 17,787
Radiation Shield + Contingency Reserve (covers price movements and TBD item)				PKR 2,213
TOTAL BUDGET REQUESTED				PKR 20,000

The radiation shield unit price is pending a local quotation from an Islamabad hardware supplier. The buffer between the hardware subtotal (PKR 17,787) and the requested total (PKR 20,000) is specifically reserved to cover this item and any minor price movements at point of order. Any unspent portion will be returned to the institute.

9.2 Future Phase Costs

The following costs are presented for transparency and forward planning only. No approval is requested for these items at this stage. Phase 2 requires no additional budget; it utilises the hardware deployed in Phase 1. Phase 3 costs will be the subject of a separate proposal following successful pilot completion.

Phase	Description	Estimated Cost	Status
Phase 1	Hardware procurement and rooftop deployment (this proposal)	PKR 20,000 (one-time)	REQUESTED
Phase 2	Model refinement, extended collection, pilot report	Negligible (uses Phase 1 assets)	No cost
Phase 3	Cloud pipeline, live dashboard, API integrations, SaaS launch	PKR 5,000-10,000 / month (est.)	Separate proposal post-pilot

10. Deliverables at Pilot Completion

The following formal deliverables will be produced and submitted to BIC management at the conclusion of the three-month pilot.

No.	Deliverable	Description
1	Validated PM2.5 Dataset	Three months of timestamped, quality-controlled hourly records from the BIC Islamabad rooftop: PM1, PM2.5, PM10, temperature, humidity, pressure, and rain-event flags.
2	Trained Forecasting Models	Validated ML models for 1-hour, 3-hour, 6-hour, and 24-hour PM2.5 prediction horizons. Documented RMSE, MAE, and R2 performance metrics per horizon.
3	Decision Framework Assessment	Pilot-mode evaluation of the operational decision layer: accuracy of triggered recommendations, cases of alignment with observed conditions, and identified failure modes.
4	SHAP Feature Analysis	Interpretability report identifying which environmental and temporal variables most strongly drive PM2.5 forecasts under Islamabad campus conditions.
5	Pilot Completion Report	A formal technical report covering system performance, data quality, model evaluation, infrastructure lessons, Karachi comparison, and a Phase 2 roadmap. Submitted to BIC management and academic supervisors.

11. Conclusion and Request for Approval

AirSense is a student-developed system with a clear architecture, a rigorous methodology, a two-city deployment network, and a defined path from campus pilot to real-world product. It was recognised in an international academic innovation context for its design and its ambition. The BIC campus pilot is the step that converts that recognition into demonstrated, measurable, real-world performance.

The ask is genuinely modest relative to what it produces. PKR 20,000 is a one-time hardware expenditure. In return, BIC receives a functioning rooftop sensor, three months of live environmental data, trained AI forecasting models calibrated to this specific campus, a tested decision support framework, and a team that is accountable to the institute for the quality of every deliverable.

The team respectfully requests approval of the following from Ms. Saba Ahson and Ms. Sahifa Alam:

1	Approval of a one-time hardware budget of PKR 20,000 for procurement of the sensor assembly and all supporting components as itemised in Section 9.
2	Permission to install the sensor assembly at an agreed rooftop location on the BIC Islamabad campus for a period of three months.
3	Facilitation of access to a rooftop power point and support for positioning of a campus Wi-Fi extender to ensure adequate connectivity to the sensor unit.
4	Recognition of this initiative as an officially supported BIC student project, enabling appropriate access to campus computing resources for data processing and model training.

The sensor goes on the roof. The data comes in. The models learn.

We ask for PKR 20,000 and three months to prove that it works.

Everything else follows from that.

Respectfully submitted,

Muhamm
ad Munim
Qureshi

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Muhammad Munim Qureshi
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Qureshi, gn=Muhammad
Munim Qureshi, c=United
States, l=US, o=BIC, ou=AI
e=munim.351816@bic.edu.pk
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Location:
Date: 2026-07-29
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Munim Qureshi

AirSense Project Team

Beaconhouse International College, Islamabad

July 2026

Approval Sign-Off

Role	Name	Signature	Date
Head of Institute	Ms. Saba Ahson		
Head of CSSE / AI	Ms. Sahifa Alam		
Project Team Lead	Munim Qureshi	<i>Munim</i>	29/07/2026